TOCI-II (Modeling and Simulation)

(Course Code: CSSE-607)

PROJECT PROPOSAL

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Project Description:

The objective of this project is to develop a stochastic, discrete-event simulation model of the patient service system at a diagnostic laboratory during peak operating hours in order to analyze the performance of the queuing system. The peak period under consideration is from **9:00 AM to 11:00 AM**, a two-hour busy interval. The system consists of a single service counter — the **Front Desk** — which handles all patient interactions. Patients join a single waiting line (queue) upon arrival and are served on a **First In, First Out (FIFO)** basis.

The simulation will model the random arrival of patients to the front desk queue and the service process at the desk. The focus of the study is to evaluate key performance measures such as:

- Average patient waiting time at the front desk
- Average queue length at the front desk
- Server utilization (front desk staff)
- Throughput (number of patients served during the peak period)

The results of this simulation will help in understanding the behavior of the laboratory service system under heavy demand and can be used to assess whether the current service capacity is sufficient or if additional staff or desks/counters are required.

System Description:

1. System Boundary

The system begins when a patient joins the waiting line at the front desk and ends when the patient completes service and departs from the laboratory service area.

The following components are included within the system boundary:

- Front desk waiting line
- Front desk service counter
- Service process at the front desk
- Patient departure after service completion

2. Entities

The **primary entity** in the system is the **Patient**, who arrives at the laboratory, joins the front desk queue, receives service at the counter, and then exits the system.

The **secondary entity** is the **Front Desk Staff (Server)**, who provides service to patients at the front desk.

Queue and Server Configuration:

- One queue: Front Desk Queue
- One service counter: Front Desk Server
- Queue discipline: First In, First Out (FIFO)

3. Arrival Process

Patients arrive to the system by joining the front desk queue directly. The arrival process is stochastic and interarrival times are modeled using a random distribution (e.g., exponential distribution). Upon completing service at the front desk, the patient departs the system.

4. Service Process

Service at the front desk encompasses all interactions between the patient and the staff member. Service times are stochastic and modeled using an appropriate probability distribution.

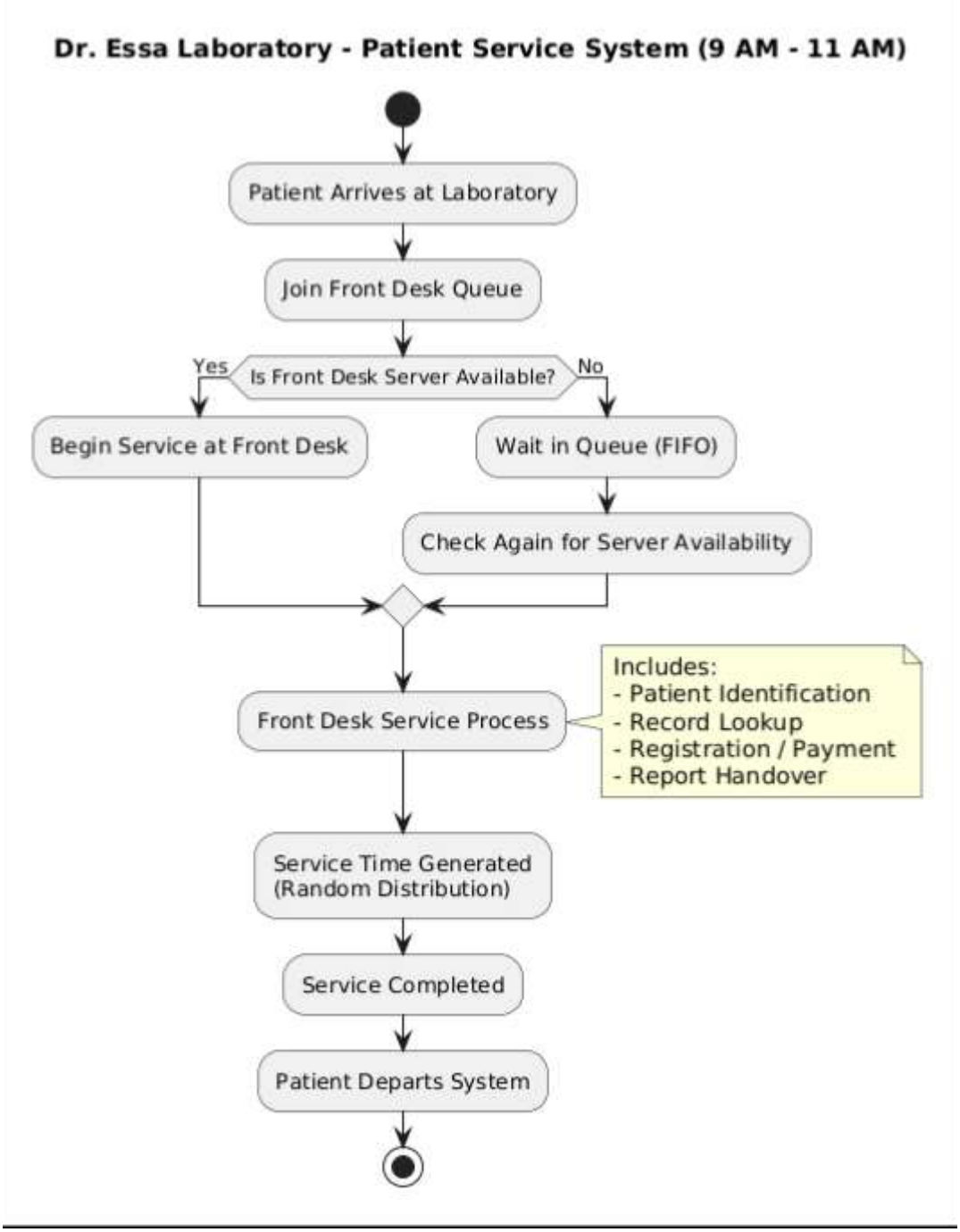
The service process includes:

- Patient identification and record lookup
- Administrative processing (registration, payment, or report handover as applicable)
- Completion of the service interaction and patient departure

All patients are treated as a single homogeneous entity class, and variability in service duration is captured entirely through the stochastic service time distribution.

Generalized Process Flow Analysis:

The patient service process follows a simple, linear flow. All patients enter the system through a single entry point and follow an identical path through the system:

1. **Patient Arrival:** The patient arrives and joins the end of the front desk queue.
2. **Wait in Queue:** The patient waits until the front desk server becomes available (FIFO discipline).
3. **Front Desk Service:** The patient is served at the front desk. Service time is drawn from a stochastic distribution.
4. **Departure:** Upon completion of service, the patient exits the system.

Process Flow Diagram (Flowchart)